Security-Constrained Economic Dispatch

Optimising the Operational Cost of a Campus Micro-grid

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1. Problem

1.1 The Shift to Time-of-Day (ToD) Pricing

Historically, electricity billing in India has relied on "Slab Pricing", where consumers pay a fixed rate that only increases if consumption crosses certain thresholds. For e.g., for residential areas : 0-200 units : zero charge, 201-400 units : Rs. 4.5 - 5/unit, 401-800 units : Rs 5-6.5/unit, etc. While common for households, it is outdated for large institutional micro-grids. To prevent massive facilities from overwhelming the city grid during evening peak hours, modern smart grids mandate **Time-of-Day (ToD) Pricing**. Under ToD, electricity becomes heavily discounted during the night and severely penalized during the evening. Optimizing the electricity usage cost while mastering time of day pricing in electricity dispatch is the future of electricity management. We can see an example of the soon to be prevelant Time-Of-Day Pricing in this news article by Times of India : https://timesofindia.indiatimes.com/city/patna/new-rules-power-consumers-to-pay-more-during-peak-hours/articleshow/129810012.cms

1.2 The Challenge

The campus still has to buy power from the main city grid, but only when either the price is low or the demand is high enough to buy high priced electricity .Time-of-Use (TOU) pricing means it is cheap late at night but extremely expensive during peak evening hours. If a facility just uses power as it comes, they end up wasting free solar energy during the day and paying premium prices for grid power at night. Because of ToD pricing and the intermittent nature of solar power, a campus without storage wastes free solar energy at noon and pays massive financial and environmental penalties at 7:00 PM.

1.3 The Objective

Our objective is to introduce a Battery Energy Storage System (BESS) and build a linear programming algorithm to **minimize the overall operational cost**. The algorithm must mathematically decide when to charge (when power is cheap) and discharge (when power is expensive), while also ensuring the campus never loses power.

2. Plan

We treat the daily energy cycle as a constrained linear programming (LP) problem. We break a full day into 24 hourly intervals ($t = 1, \ldots, 24$) and use optimization to find the exact cost-minimizing schedule.

2.1 Decision Variables

For every hour t, the algorithm needs to decide how to dispatch the electricity. We define the following variables :

- s_t: Power used from Solar panels (in kW).
- w_t: Power used from Wind turbines (in kW).

- g_t: Power bought from the main city grid (in kW).
- b_t^{in}: Power pushed into the battery to charge it (in kW).
- b_t^{out}: Power pulled out of the battery to discharge it (in kW).
- E_t: The State of Charge (how much energy is sitting in the battery at the end of the hour t) (in kWh).

2.2 Objective Function

Our objective is to keep the total daily cost (Z) as low as possible. This includes the minor maintenance costs of renewable sources, the hourly fluctuating price of the city grid, and the degradation cost of wearing out the battery.

$$\text{Minimize } Z = \sum_{t=1}^{24} \left(C_s s_t + C_w w_t + C_{g,t} g_t + C_b (b_t^{in} + b_t^{out}) \right) \tag{1}$$

Where $C_s, C_w, C_{g,t}$, and C_b are the real-world prices per kW for each source.

2.3 System Parameters (Constants)

Unlike decision variables, these are the fixed numbers provided by our hardware and historical data, which will be useful in defining our constraints:

- $L_t, S_t^{max}, W_t^{max}$: The predicted hourly demand, maximum available solar power at hour t, and maximum available wind power at hour t (in kW).
- E_{max}: The maximum physical capacity of the battery (in kWh).
- η: The round-trip efficiency of the battery (accounting for heat loss).
- E_0: The initial energy stored in the battery at the beginning of the day (Midnight). (assumed to be 0 because we assume that all the battery energy must have been used during the peak hours at night.)

2.4 System Constraints

We restrict the optimization with the following constraints:

- **Stochastic Load Balancing:** To account for unpredictable human behavior (e.g., unexpected mass HVAC(heating,ventilarion and air-conditioning) usage), we introduce a random noise multiplier, $\epsilon_t \sim \mathcal{U}(1.0, 1.20)$), into the baseline campus demand (L_t). This simulates random demand spikes of up to 20%. The total power supplied minus power stored must equal this new randomized demand ($\tilde{L}_t$) every hour.

$$\tilde{L}_t = L_t \cdot \epsilon_t \tag{2}$$

$$s_t + w_t + g_t + b_t^{out} - b_t^{in} = \tilde{L}_t \quad \forall t \in [1, 24] \tag{3}$$

A note on our random noise generator (ϵ_t), By setting the lower limit to 1.0, we ensured the algorithm never received an unrealistic drop in demand—every hour

was either exactly as forecasted, or worse. We capped the maximum at 1.2 because a 20% unexpected load surge represents the standard engineering safety margin for severe events, like an unforecasted heatwave. Finally, we chose a Uniform distribution rather than a Normal distribution because we wanted maximum volatility; we wanted a brutal 20% spike to be just as likely as a 2% spike to truly push the battery's state-of-charge to its limits

- **Generation limits:** We cannot pull more solar or wind power than the weather actually provides

$$s_t \leq S_t^{max} \quad \text{and} \quad w_t \leq W_t^{max} \quad \forall t \tag{4}$$

- **Battery Physics (State of Charge):** A battery stores energy. The energy inside it right now depends on what was in it an hour ago, minus energy lost to heat during charging (η). Thus, This constraint makes our optimization problem inter-dependent, i.e. energy dispatch at hour t is wholly dependent on any other hour of the day, to optimize the operational cost.
Note: For $t = 1$, E_{t-1} is defined as our parameter E_0 (0).

$$E_t = E_{t-1} + (\eta \cdot b_t^{in} \cdot \Delta t) - \left(\frac{b_t^{out}}{\eta} \cdot \Delta t\right) \quad \forall t, (\eta \in (0, 1]) \tag{5}$$

$$\text{As } \Delta t = 1 \ \forall t,$$

$$E_t = E_{t-1} + (\eta \cdot b_t^{in}) - \left(\frac{b_t^{out}}{\eta}\right) \quad \forall t, (\eta \in (0, 1]) \tag{6}$$

- **Capacity Constraint:** We cannot overcharge the battery past its physical limit (E_{max}), and we cannot drain it below zero.

$$E_t \leq E_{max} \quad \forall t \tag{7}$$

- **Non-negativity Constraint:** We cannot drain the battery energy below zero and we cannot demand negative solar and wind power.

$$s_t, w_t, E_t \geq 0 \tag{8}$$

3. Data

3.1 The Case for Synthetic Data

Obtaining perfectly clean, localized data for a specific campus is notoriously difficult due to proprietary smart-meter restrictions. Consequently, we constructed a 24-hour synthetic dataset. While synthetic data lacks the micro-complexities of a physical location, it provides a **Controlled Environment**. It also helps to stress-test the battery logic under extreme price fluctuations. Also, to somewhat overcome our limitation of choosing synthetic data over real-time data, we added a random noise generator, introduced in 2.4.

3.2 System Parameters (The Constants)

We assume the campus micro-grid is equipped with a standard battery system. These are the fixed starting parameters we feed into the model:

- **Battery Capacity (E_{max}):** 250 kWh (The physical limit of the system).
- **Charging Efficiency (η):** 95% or 0.95 (Accounting for minor energy lost to heat during transfer).
- **Initial State (E_0):** 0 kWh (We assume the battery is completely depleted at midnight, $t = 0$, establishing a strict baseline).

3.3 Fixed Operational Costs

While renewable energy feels "free", the hardware requires maintenance. Furthermore, using a battery slowly degrades its chemical lifespan. We define our fixed costs as:

- **Solar (C_s) & Wind (C_w):** Rs. 0.5 per kWh (Cost of cleaning and mechanical upkeep, 0.5 as a real life based assumption, can be calculated using this formula, cost of maintenance per unit = yearly cost received to maintain(solar panels/wind turbines) / yearly units of energy produced).
- **Battery Cycling (C_b):** Rs. 2.0 per kWh (The chemical degradation cost per cycle, also can be calculated by the formula, degradation cost of battery = price of battery/total energy storage capacity in lifetime (in cycles promised)).

3.4 Time-Series Data (The Hourly Variables)

The dynamic data for our 24-hour horizon is structured to force the algorithm to make difficult optimization trade-offs (These are values from our synthetic data which mirror the real life situations):

- **Grid Price ($C_{g,t}$):** Follows a strict Time-of-Use (TOU) tariff. Electricity is cheap during off-peak night hours (Rs. 4.0/kWh) and heavily penalized during the evening peak (Rs. 15.0/kWh).
- **Solar Generation (S_t^{max}):** Peaks sharply at noon (up to 120 kW) but drops completely to zero by 6:00 PM.
- **Campus Demand (L_t):** Remains moderate during the working day but spikes to 200 kW between 7:00 PM and 9:00 PM (exactly when solar is unavailable and grid prices are highest).

4. Analysis (Implementation & Results)

4.1 Model Implementation

To compute the optimal 24-hour dispatch schedule, we constructed our linear programming model using the `lpSolve` package in the R programming environment. The algorithm was tasked with finding the absolute minimum value for our objective function

across 144 interconnected decision variables (6 variables per hour $\times$ 24 hours), strictly bound by our load-balancing and battery physics constraints.

Note: The full, reproducible R script and the dataset are provided in the github link mentioned below.

GitHub Repository: https://github.com/joshimaharshi18/energy-grid-optimization

4.2 Key Findings & Battery Behavior

Upon running the optimization, the model successfully satisfied 100% of the campus power demand while actively avoiding premium grid prices. The mathematical solver converged on a highly strategic schedule, demonstrating three distinct operational phases:

- **Phase 1: Midday (Peak of Solar Power):** Between 10:00 AM and 3:00 PM, solar generation exceeded campus demand. Instead of wasting this free energy, the algorithm aggressively routed the excess power into the battery (b_t^{in}), absorbing all the power not being used for the hourly demand to charge the battery, preparing for peak hours and grid surges.
- **Phase 2: Evening (No solar,Peak grid):** Exactly at 6:00 PM, solar generation dropped to zero, demand spiked to 200 kW, and the grid price surged to Rs. 15.0/kWh. The algorithm immediately dropped grid purchasing (g_t) to the absolute minimum required to optimize the cost. It relied heavily on discharging the battery (b_t^{out}) to carry the campus through the expensive peak hours, but also keeping it discharged until the peak hours get finished.
- **Phase 3: Night (Off-peak grid):** After 10:00 PM, when the campus demand dropped and the grid price returned to a cheap Rs. 4.0/kWh, the algorithm utilized the grid not only to run the campus but to pre-charge the battery to keep it ready for the next day's surge.

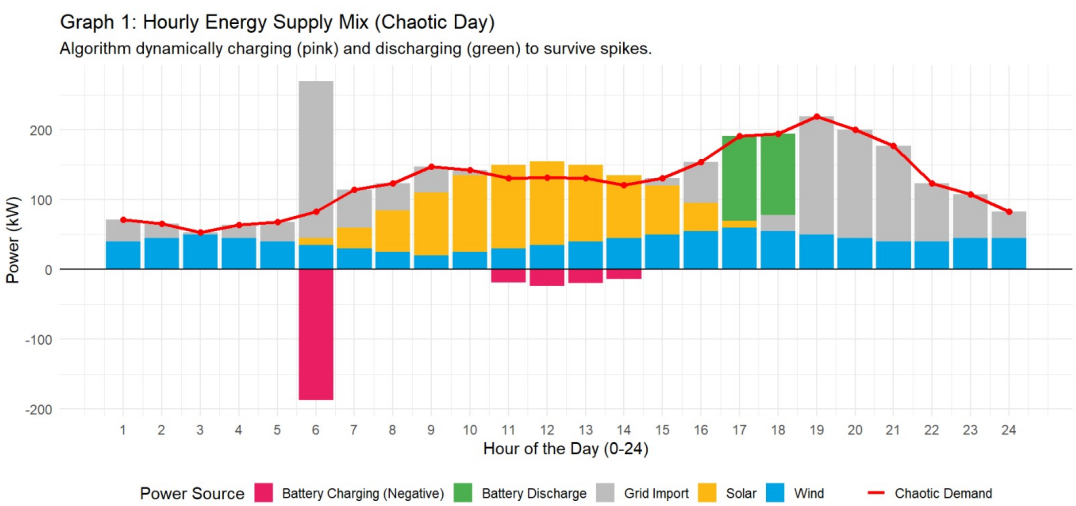

Figure 1: Hourly Energy Supply Mix (Chaotic Day). The algorithm dynamically charges (pink) and discharges (green) to survive spikes.

5. Conclusion

5.1 Summary of Savings & ROI Projections

This project successfully demonstrates that a Battery Energy Storage System (BESS), when managed by an intelligent optimization algorithm, is a highly effective financial tool for a campus micro-grid. By treating the daily energy cycle as a linear programming problem, we proved that facilities can mathematically bypass the financial penalties of Time-of-Use pricing and grid surges.

- **Baseline Daily Cost :** Rs. 15009.47 (when all energy comes from the grid)
- **Optimized Daily Cost :** Rs. 13234.79 (when our optimization model is at work)
- **Total Daily Savings: Rs. 1774.68**

After a little bit of repeated simulations, we gathered 20 daily savings values as a little sample to calculate and estimate the **Annual Savings** our optimization model maybe giving, which is coming out to be **Rs. 655697.315** [365 * (mean of values - (1824.05, 1792.62, 1770.83, 1762.08, 1801.28, 1808.13, 1851.81, 1830.88, 1764.7, 1769.81, 1785.24, 1759.27, 1896.64, 1758.56, 1761.08, 1788.68, 1758.2, 1747.85, 1892.38, 1804.54))]

5.2 Robustness Testing (Randomized Stress Test) :

To ensure our model survives real-world chaos, we injected a uniform random multiplier ($\epsilon_t \sim \mathcal{U}(1.0, 1.20)$) to simulate a "Worst-Case Day" where unexpected events cause demand to randomly surge by up to 20% (Figure 2).

As detailed in Table 1, the algorithm dynamically adapted to the new chaotic load ($\tilde{L}_t$) without causing load-shedding. Most notably, the data reveals *the battery acts as an insurance policy.* On a chaotic day, a dumb grid absorbs massive spikes at the brutal Rs. 15 peak rate. Because our algorithm adapted, the comparative financial savings actually increased during the worst-case scenario.

5.3 Limitations :

As any project would have some, Our project also has some limitations.

- **Synthetic data:** We could not find real data which would have made our project real-life based simulation, but our synthetic data was made by keeping in mind the real life needs and energy surges hourly.
- **Prediction of energy demand:** To start the day at hour 1 with optimized energy dispatch schedule, we have to predict next day's energy demand and feed it into the model which will then give us the full schedule of the next day's energy dispatch hourly cycle. Our random noise generator helps with the uncertainty that energy consumption can surge anytime.

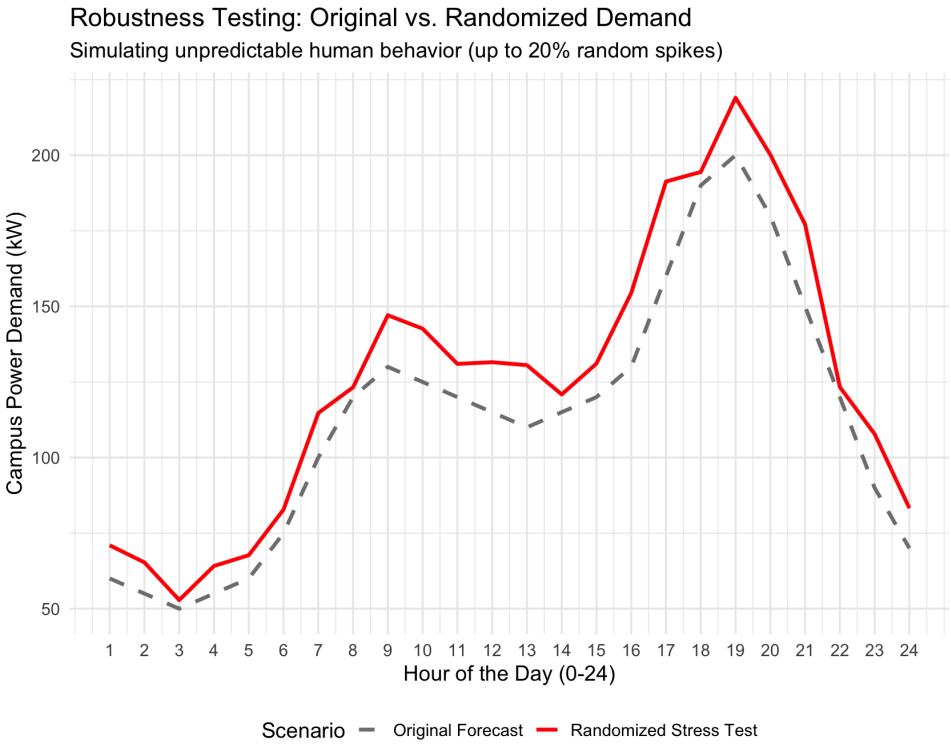

Figure 2: Stress Test Scenario: The original perfect demand forecast versus the randomized chaotic demand curve ($\tilde{L}_t$).

Table 1: Optimized Dispatch Schedule for the Randomized Chaotic Day

Hour	$\tilde{L}_t$ (Load)	Solar Used	Wind Used	Grid Bought	Batt Charge	Batt Disch	Batt Level	Grid Price
1	65.5	0.0	40.0	25.5	0.0	0.0	0.0	4
2	60.1	0.0	45.0	15.1	0.0	0.0	0.0	4
3	52.6	0.0	50.0	2.6	0.0	0.0	0.0	4
4	62.4	0.0	45.0	17.4	0.0	0.0	0.0	4
5	67.2	0.0	40.0	288.7	261.4	0.0	248.4	4
6	80.6	10.0	35.0	37.3	1.7	0.0	250.0	4
7	113.8	30.0	30.0	53.8	0.0	0.0	250.0	8
8	132.8	60.0	25.0	47.8	0.0	0.0	250.0	8
9	145.4	90.0	20.0	35.4	0.0	0.0	250.0	8
10	138.8	110.0	25.0	3.8	0.0	0.0	250.0	8
11	124.6	120.0	30.0	0.0	25.4	0.0	250.0	8
12	132.9	120.0	35.0	0.0	22.1	0.0	250.0	8
13	129.4	110.0	40.0	0.0	20.6	0.0	250.0	8
14	131.0	90.0	45.0	0.0	4.0	0.0	250.0	8
15	129.5	70.0	50.0	9.5	0.0	0.0	250.0	8
16	133.2	40.0	55.0	38.2	0.0	0.0	250.0	8
17	190.2	10.0	60.0	120.2	0.0	0.0	250.0	15
18	192.1	0.0	55.0	0.0	0.0	137.1	105.7	15
19	229.4	0.0	50.0	79.0	0.0	100.4	0.0	15
20	197.6	0.0	45.0	152.6	0.0	0.0	0.0	15
21	175.7	0.0	40.0	135.7	0.0	0.0	0.0	15
22	123.6	0.0	40.0	83.6	0.0	0.0	0.0	8
23	92.5	0.0	45.0	47.5	0.0	0.0	0.0	4
24	83.5	0.0	45.0	38.5	0.0	0.0	0.0	4

Team Contributions :

There were equal contributions in value by all three team members. The members researched their parts and wrote the latex code as well.

- Maharshi Joshi (BSDDH2511) took initiative for the PROBLEM and PLAN sections.
- Vansh Makhija (BSDDH2523) worked on the DATA and ANALYSIS sections
- Ketan Vishnoi (BSDDH2509) handled the CONCLUSION section and led the team in the writing the R CODE for the optimisation model

We divided parts of the report amongst ourselves but the whole report, research and the R code for the model took all of our help and dedication which we did together.